measurement of the other. Consequently, when one magnitude is ascertained with absolute accuracy, the other one remains absolutely indefinite.

It is evident that this principle fundamentally precludes the possibility of translating into the sense world, with an arbitrary degree of accuracy, the simultaneous values of the coordinates and momenta of material points, as these are conceived in the world picture of classical physics; this circumstance constitutes a difficulty with respect to the recognition of a universal validity of the principle of strict causality, and it has even caused some indeterminists to regard the law of causality in physics as decisively refuted. However, upon closer scrutiny, this conclusion—founded on a confusion of the world picture with the sense world—proves a rash one, to say the very least. For there is another, more logical way out of the difficulty, a way which has often rendered excellent services on previous occasions—namely, the assumption that the attempt to determine simultaneously both the coordinates and the momentum of a material point is physically completely meaningless. However, the impossibility of giving an answer to a meaningless question must, of course, not be charged up against the law of causality, but solely against the premises which produced that particular question; in other words, in this particular instance, against the assumed structure of the world picture of physics. And since the classical world picture has failed, another must take its place.

This is what actually has happened. The new world picture of quantum mechanics is a product of the need to find a way of reconciling the quantum of action with the principle of strict determinism. For this purpose, the traditional primary constituent of the world picture, the material point, had to be deprived of its basic, elementary character; it was resolved into a system of material waves. These material waves constitute the primary elements of the new world picture. The material point in its old meaning now appears merely as a special borderline case, as an infinitesimally small parcel of waves, the momentum of which is totally indefinite, since its position is definite, according to Heisenberg’s uncertainty principle. If we assign a certain range to the position of the material point, the value of the momentum also becomes approximately definite and the laws of classical mechanics then are approximately valid for positions and momenta.

In general, the laws of material waves are basically different from those of the classical mechanics of material points. However, the point of central importance is that the function characterising material waves, i.e. the wave function or probability function (the term itself is of no importance here), is fully determined for all places and times by the initial and boundary conditions, according to quite definite principles of computation, whether we use Schroedinger’s operators, Heisenberg’s matrices, or Dirac’s *q*-numbers.

We thus see that the principle of determinism is as strictly valid in the